

Cellular Immunity in Leprosy

Lesions in lepromatous leprosy are painful but not tender. The patient has **Lucio's phenomenon** (a severe form of lepromatous leprosy), manifested by: - Diffuse skin infiltration - Total alopecia of the eyelashes and eyebrows - Nasal septum perforation - Absence of nodular lesions - Severe stocking-glove pattern of sensory impairment

The study of leprosy provides critical insights into: 1. Understanding the immunopathogenesis of the granulomatous spectrum and reactional states
2. Eradication or control of the disease through vaccination
3. Use of leprosy as a model to probe **cell-mediated immunity (CMI)** in humans, which can inform understanding of other diseases

Polar Forms of Leprosy

Leprosy exhibits a wide clinical and immunological spectrum, best illustrated by its two polar forms:

- **Tuberculoid leprosy (TT)** – high resistance form: - Few well-defined skin lesions - Rare acid-fast bacilli - Epithelioid granulomas - Strong cell-mediated immunity - Tendency toward self-cure
- **Lepromatous leprosy (LL)** – low resistance form: - Widespread dissemination - Abundant acid-fast bacilli - Foamy macrophages - Poor cell-mediated immunity - Progressive disease if untreated

A sharply demarcated plaque reflects delayed-type hypersensitivity (DTH) to *M. leprae* in the skin; nerve trunk palsy represents DTH activity in peripheral nerves.

Evidence for Cellular Immunity in Host Defense

- **Leprimin skin test:** Positive in tuberculoid patients, negative in lepromatous patients — first objective evidence that host cellular immunity determines disease polarity.

- **Lymphocyte transformation tests:** Correlate with leprimin results, confirming that cellular immunity mediates polar differences.
- **Humoral immunity:** Antibody responses are stronger in LL patients, but do not confer protection — indicating that antibodies are not protective against *M. leprae*.

Lymphocyte Subsets in Skin Lesions

Immunophenotyping reveals distinct T-cell infiltration patterns: - **Tuberculoid lesions:** Predominance of **CD4+ T cells** (CD4/CD8 ratio = 2:1) - **Lepromatous lesions:** Predominance of **CD8+ T cells** (CD4/CD8 ratio = 1:2)

Notably, peripheral blood shows normal CD4/CD8 ratios (2:1) in both forms, emphasizing the importance of analyzing immune responses at the **site of disease activity** — the skin lesions.

Cytokine Profiles in Polar Lesions

mRNA analysis via reverse transcriptase-polymerase chain reaction (RT-PCR) reveals functionally distinct cytokine environments:

- **Tuberculoid lesions (Th1 profile):**
 - High expression of:
 - Interleukin-2 (IL-2)
 - Interferon- γ (IFN- γ)
 - IL-12
 - Low expression of:
 - IL-4
 - IL-10
 - CD4+ T cells produce IFN- γ → activates macrophages and promotes CMI
- **Lepromatous lesions (Th2 profile):**
 - High expression of:
 - IL-4

- IL-10
- Low expression of Th1 cytokines
- CD8⁺ T cells are primary source of IL-4

Functional Consequences

- **Th1 response (TT):** Promotes strong T-cell and macrophage activation → effective containment of infection.
- **Th2 response (LL):** Drives antibody production but suppresses CMI → uncontrolled bacterial proliferation and disease progression.

Therapeutic Implications

- Intradermal administration of recombinant **IFN- γ** in lepromatous patients reduces bacillary load, demonstrating the potential to boost CMI therapeutically.

Innate Immunity and Immune Polarization

The host's decision to mount a Th1 or Th2 response may be influenced by the **innate immune system's recognition of *M. leprae***, setting the stage for downstream adaptive immune polarization.